

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	26453	48.605395
1	[Git, Commit]	6754	12.409966
2	[Git, Change, Create]	5851	10.750772
3	[Git, Change, Delete]	4197	7.711671
4	[File, Git]	3579	6.576143
5	[Git, Change, Rename]	1699	3.121785
6	[Author, Git, Person]	1000	1.837425
7	[Git, Tag]	916	1.683081
8	[Json, Key]	668	1.227400
9	[Json, Value, Scalar]	603	1.107967
10	[Committer, Git, Person]	356	0.654123
11	[NPM, Dependency]	338	0.621050
12	[Type, TS, Declared]	294	0.540203
13	[Type, TS, Primitive]	294	0.540203
14	[TS, ExternalDeclaration]	215	0.395046
15	[Type, TS, Literal]	138	0.253565
16	[Json, Value, Object]	133	0.244377
17	[Type, TS, Union]	121	0.222328
18	[Type, TS, ObjectMember]	111	0.203954
19	[NPM, Script]	91	0.167206
20	[TS, Property]	65	0.119433
21	[TS, Function]	47	0.086359
22	[TS, Parameter]	44	0.080847
23	[Type, Object, TS]	42	0.077172
24	[Type, TS, FunctionParameter]	40	0.073497
25	[File, Directory]	34	0.062472
26	[Type, TS, Function]	34	0.062472
27	[Package, File, Json, NPM]	29	0.053285
28	[TS, Variable]	24	0.044098
29	[Value, TS, Literal]	20	0.036748

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

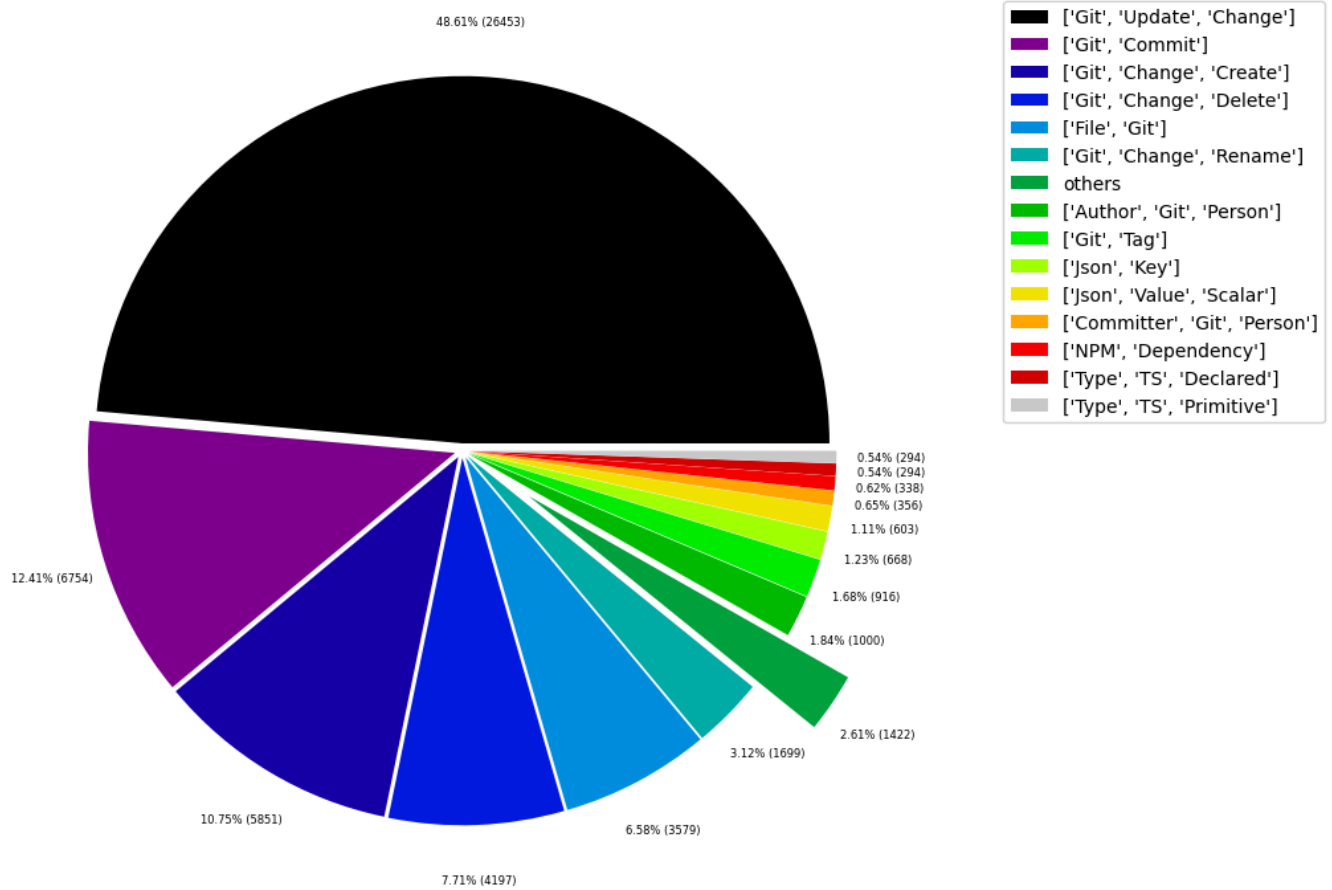


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.001837
1	[NPM, PublishConfig]	1	0.001837
2	[Value, TS, ObjectMember]	1	0.001837
3	[TS, Class]	1	0.001837
4	[Git, Branch]	1	0.001837
5	[TS, Method]	1	0.001837
6	[Repository, File, Git]	1	0.001837
7	[TS, Constructor]	1	0.001837
8	[TS, Enum]	2	0.003675
9	[Value, Object, TS]	3	0.005512
10	[Type, TS, Tuple]	3	0.005512
11	[Value, TS, Function]	4	0.007350
12	[TS, TypeParameter]	4	0.007350
13	[Repository, NPM]	5	0.009187
14	[File, TS, Scan]	5	0.009187
15	[File, Local]	5	0.009187
16	[Value, TS, Complex]	5	0.009187
17	[Project, TS]	5	0.009187
18	[Value, TS, Member]	6	0.011025
19	[File, TS, Local, Module]	6	0.011025
20	[Value, TS, Call]	6	0.011025
21	[Type, TS, TypeParameterReference]	6	0.011025
22	[NPM, Engine]	6	0.011025
23	[TS, EnumMember]	8	0.014699
24	[TS, ExternalModule]	11	0.020212
25	[Type, TS, NotIdentified]	11	0.020212
26	[Json, Value, Array]	12	0.022049
27	[Value, TS, Declared]	13	0.023887
28	[Git, Change, Copy]	15	0.027561
29	[TS, TypeAlias]	16	0.029399

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (less than 0.5% overall)

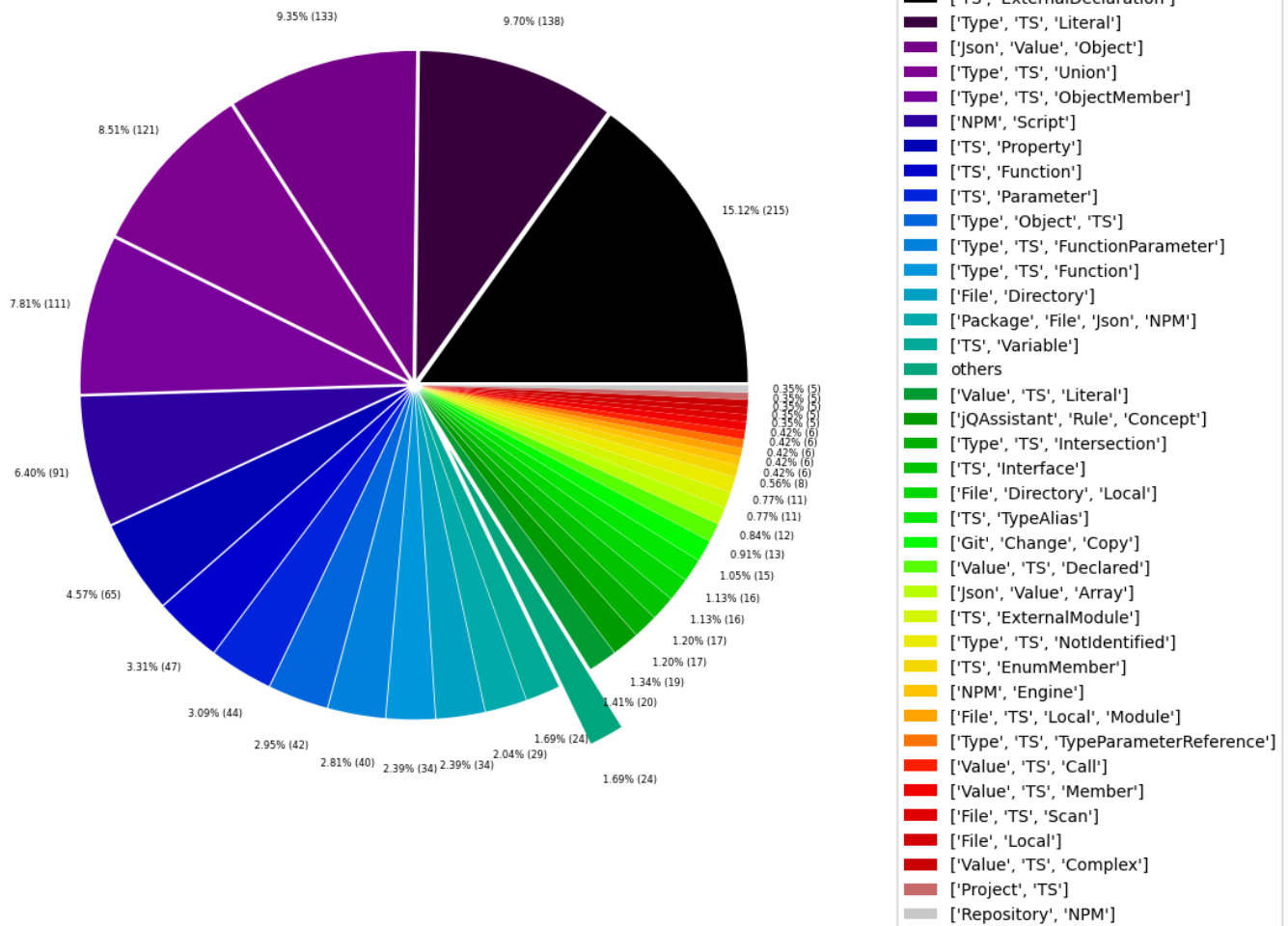


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

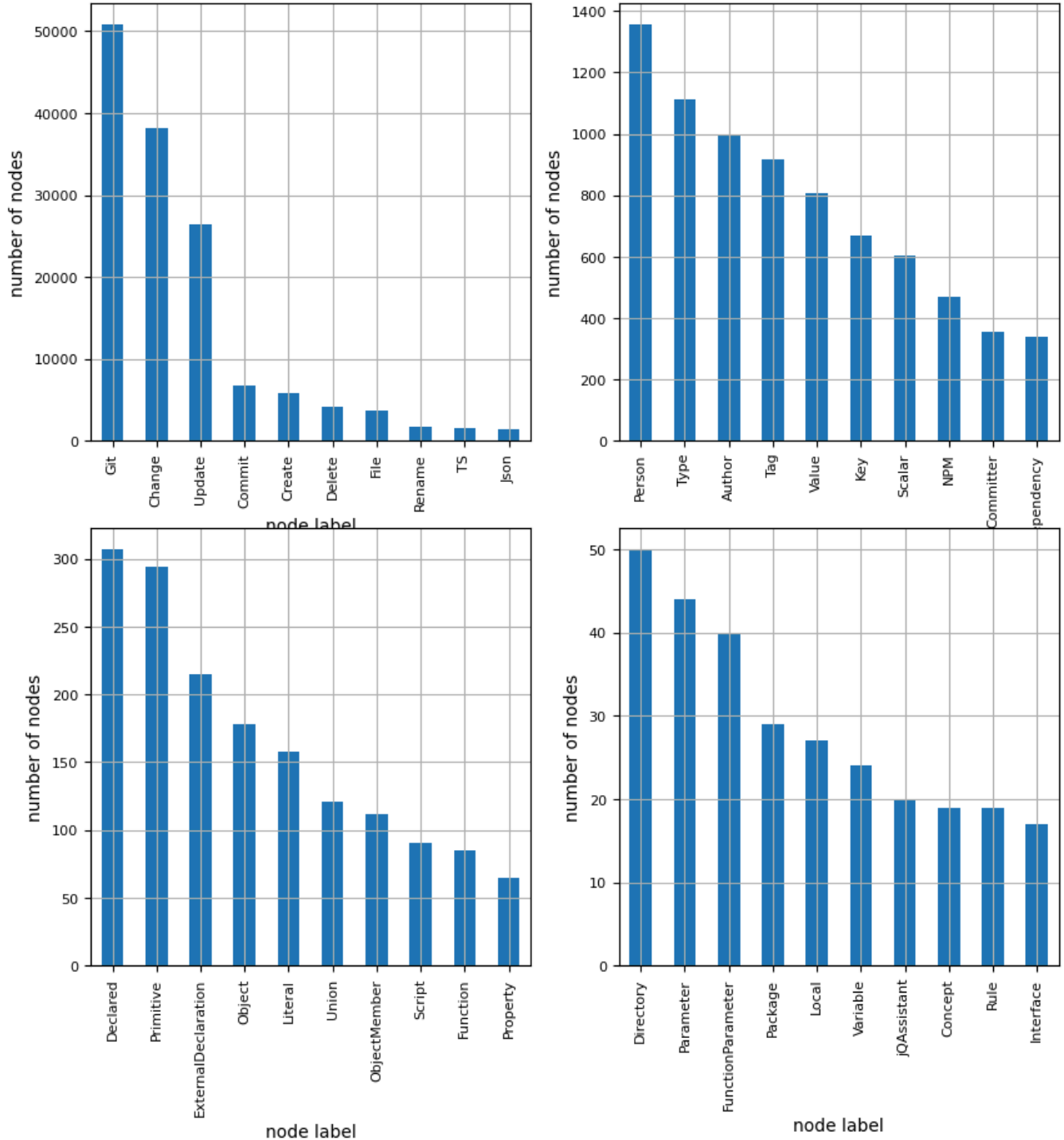
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	50822	93.381596
1	Change	38215	70.217184
2	Update	26453	48.605395
3	Commit	6754	12.409966
4	Create	5851	10.750772
5	Delete	4197	7.711671
6	File	3675	6.752536
7	Rename	1699	3.121785
8	TS	1641	3.015214
9	Json	1445	2.655079
10	Person	1356	2.491548
11	Type	1111	2.041379
12	Author	1000	1.837425
13	Tag	916	1.683081
14	Value	806	1.480964
15	Key	668	1.227400
16	Scalar	603	1.107967
17	NPM	470	0.863590
18	Committer	356	0.654123
19	Dependency	338	0.621050
20	Declared	307	0.564089
21	Primitive	294	0.540203
22	ExternalDeclaration	215	0.395046
23	Object	178	0.327062
24	Literal	158	0.290313
25	Union	121	0.222328
26	ObjectMember	112	0.205792
27	Script	91	0.167206
28	Function	85	0.156181
29	Property	65	0.119433
30	Directory	50	0.091871
31	Parameter	44	0.080847
32	FunctionParameter	40	0.073497
33	Package	29	0.053285
34	Local	27	0.049610
35	Variable	24	0.044098
36	jQAssistant	20	0.036748
37	Concept	19	0.034911
38	Rule	19	0.034911
39	Interface	17	0.031236

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

<Figure size 640x480 with 0 Axes>

Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 160367

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	38215	23.829716
1	MODIFIES	38215	23.829716
2	UPDATES	26453	16.495289
3	COMMITTED	13508	8.423179
4	CREATES	7565	4.717305
5	HAS_PARENT	7528	4.694233
6	HAS_COMMIT	6754	4.211590
7	DELETES	5896	3.676567
8	HAS_FILE	3579	2.231756
9	CHANGED_TOGETHER_WITH	1725	1.075658
10	RENAMES	1699	1.059445
11	HAS_NEW_NAME	1032	0.643524
12	HAS_AUTHOR	1000	0.623570
13	HAS_TAG	916	0.571190
14	ON_COMMIT	916	0.571190
15	DEPENDS_ON	898	0.559966
16	HAS_KEY	668	0.416545
17	HAS_VALUE	668	0.416545
18	CONTAINS	598	0.372895
19	OF_TYPE	358	0.223238
20	HAS_COMMITTER	356	0.221991
21	EXPORTS	309	0.192683
22	REFERENCES	212	0.132197
23	DECLARES	186	0.115984
24	DECLARES_DEV_DEPENDENCY	169	0.105383
25	DECLARES_DEPENDENCY	161	0.100395
26	HAS_MEMBER	112	0.069840
27	HAS_TYPE_ARGUMENT	96	0.059863
28	DECLARES_SCRIPT	91	0.056745
29	HAS_PARAMETER	84	0.052380

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

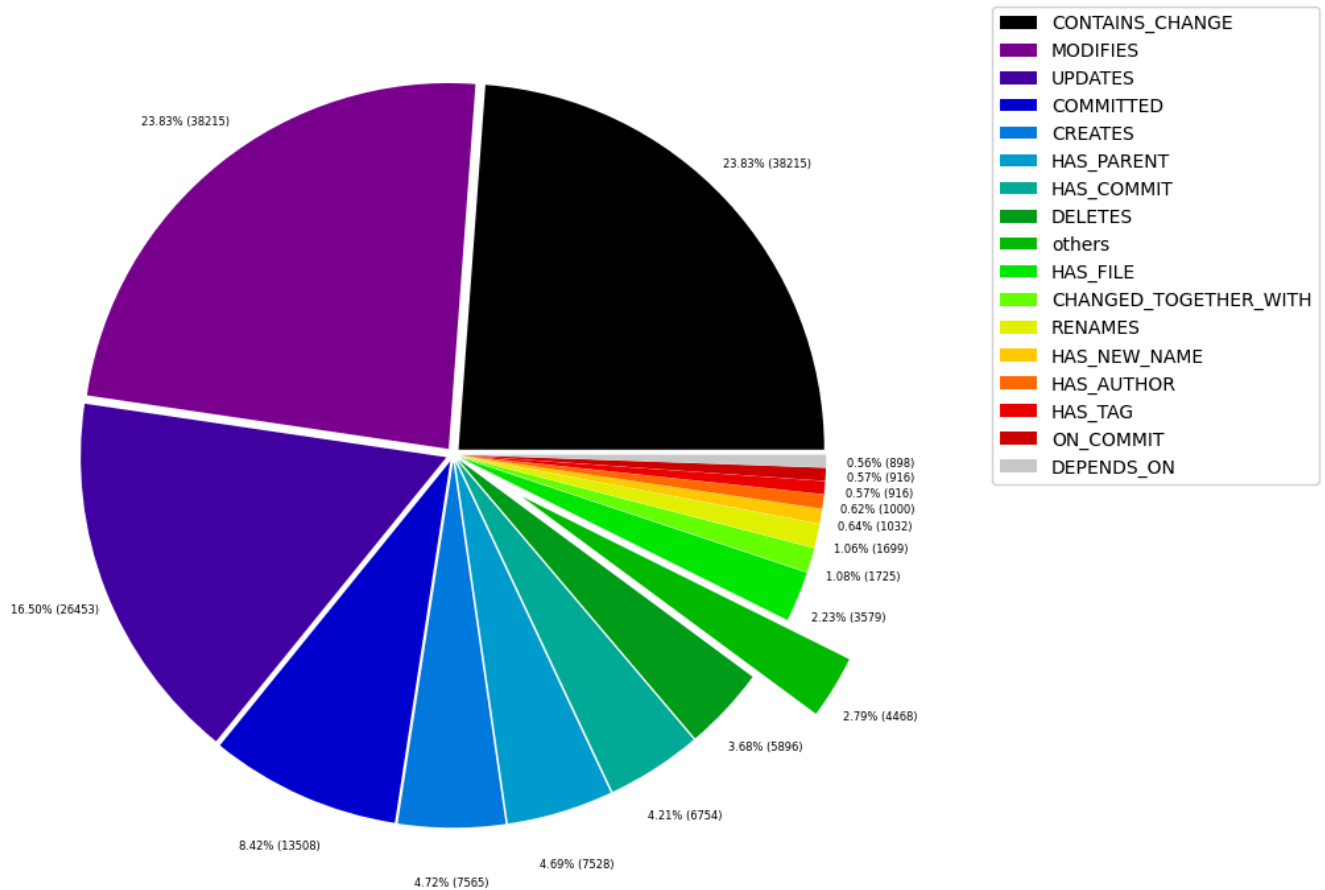


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

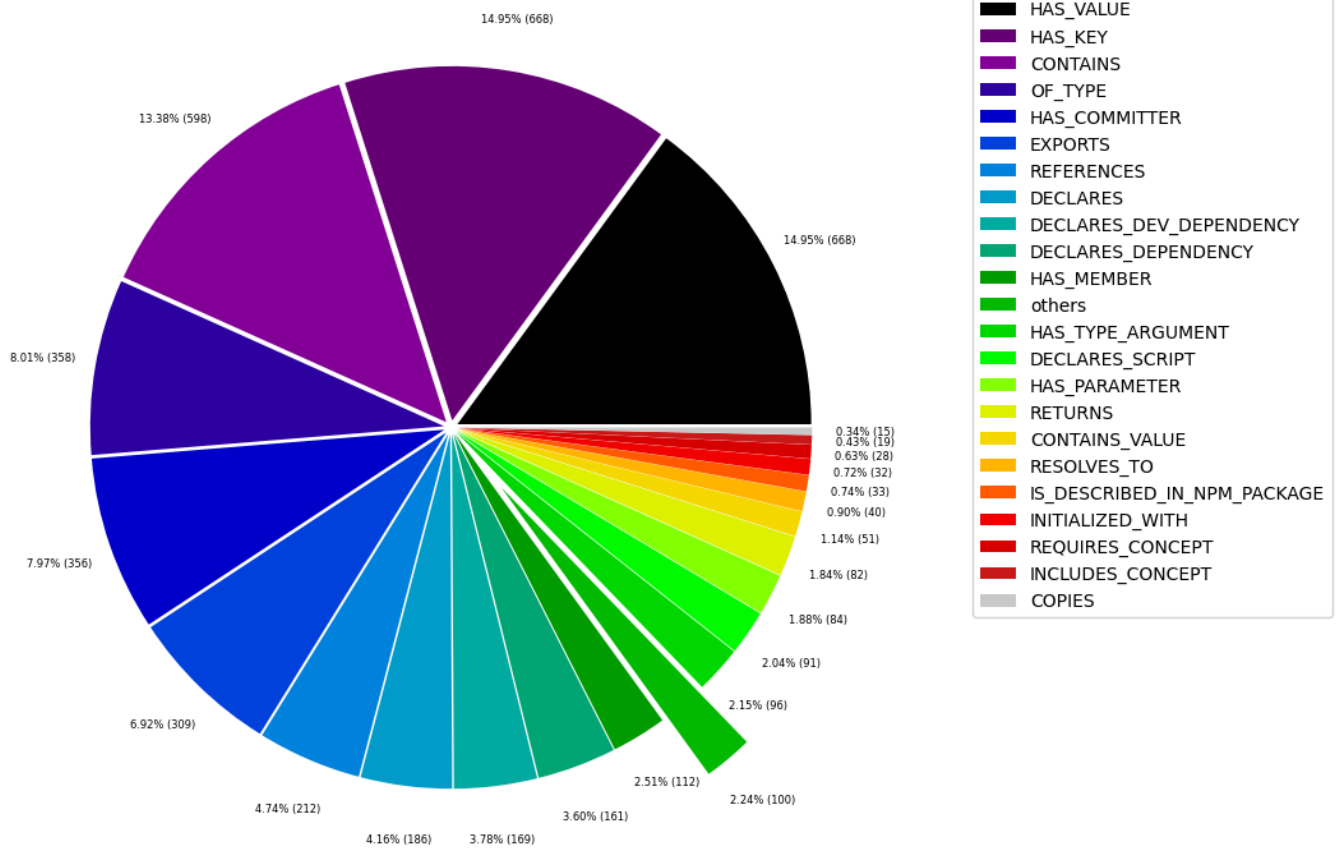
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_BRANCH	1	0.000624
1	DECLARES_PUBLISH_CONFIG	1	0.000624
2	HAS_HEAD	2	0.001247
3	CONSTRAINED_BY	4	0.002494
4	IN_REPOSITORY	5	0.003118
5	HAS_ROOT	5	0.003118
6	HAS_NPM_PACKAGE	5	0.003118
7	HAS_CONFIG	5	0.003118
8	CONTAINS_PROJECT	5	0.003118
9	DECLARES_ENGINE	6	0.003741
10	EXTENDS	6	0.003741
11	CALLS	6	0.003741
12	MEMBER	6	0.003741
13	PARENT	6	0.003741
14	HAS_ARGUMENT	6	0.003741
15	DECLARES_PEER_DEPENDENCY	8	0.004989
16	USES	11	0.006859
17	COPY_OF	12	0.007483
18	COPIES	15	0.009354
19	INCLUDES_CONCEPT	19	0.011848
20	REQUIRES_CONCEPT	28	0.017460
21	INITIALIZED_WITH	32	0.019954
22	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.020578
23	RESOLVES_TO	40	0.024943
24	CONTAINS_VALUE	51	0.031802
25	RETURNS	82	0.051133
26	HAS_PARAMETER	84	0.052380
27	DECLARES_SCRIPT	91	0.056745
28	HAS_TYPE_ARGUMENT	96	0.059863
29	HAS_MEMBER	112	0.069840

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Update, Change]	UPDATES	[File, Git]	26453	26453	
1	[Git, Update, Change]	MODIFIES	[File, Git]	26453	26453	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	26453	6754	
3	[Git, Commit]	HAS_PARENT	[Git, Commit]	7528	6754	
4	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	6754	1	
5	[Author, Git, Person]	COMMITTED	[Git, Commit]	6754	1000	
6	[Committer, Git, Person]	COMMITTED	[Git, Commit]	6754	356	
7	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	5851	6754	
8	[Git, Change, Create]	CREATES	[File, Git]	5851	5851	
9	[Git, Change, Create]	MODIFIES	[File, Git]	5851	5851	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	4197	6754	
11	[Git, Change, Delete]	DELETES	[File, Git]	4197	4197	
12	[Git, Change, Delete]	MODIFIES	[File, Git]	4197	4197	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	3579	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	1699	6754	
15	[Git, Change, Rename]	RENAMES	[File, Git]	1699	1699	
16	[Git, Change, Rename]	CREATES	[File, Git]	1699	1699	
17	[Git, Change, Rename]	DELETES	[File, Git]	1699	1699	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	1699	1699	
19	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	1578	3579	
20	[File, Git]	HAS_NEW_NAME	[File, Git]	1032	3579	
21	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1000	1	
22	[Repository, File, Git]	HAS_TAG	[Git, Tag]	916	1	
23	[Git, Tag]	ON_COMMIT	[Git, Commit]	916	916	
24	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
25	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
26	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	356	1	
27	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
28	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
29	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	

Graph Density

total_number_of_nodes (vertices): 54424

total_number_of_relationships (edges): 160367

-> total directed graph density: 5.414296921273447e-05

-> total directed graph density in percent: 0.005414296921273447